

Urban Traffic Volume Prediction Using Machine Learning Algorithms

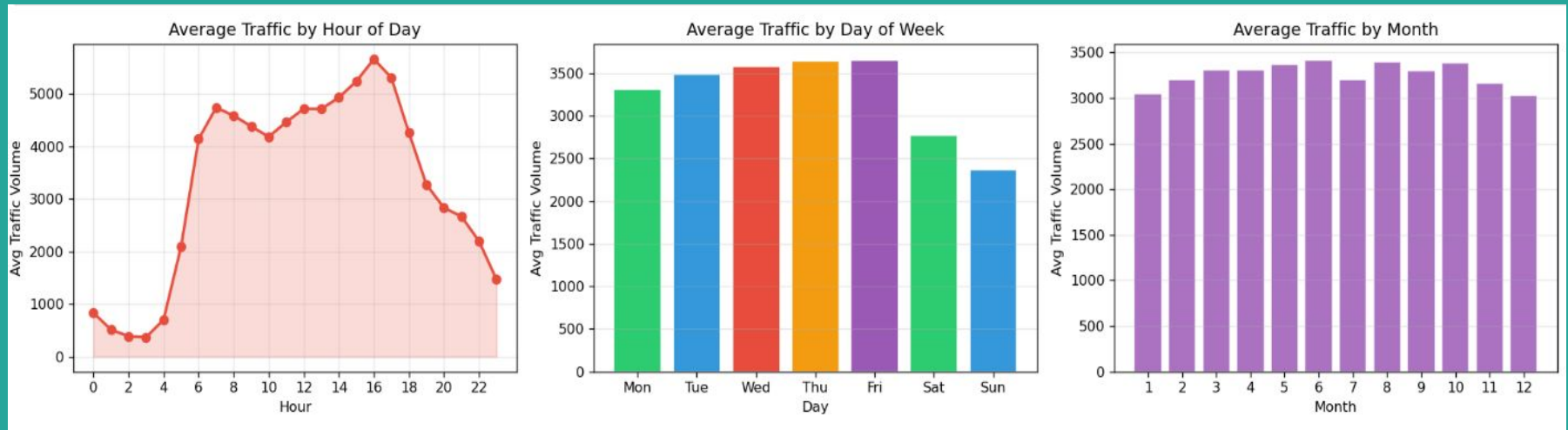
Undergraduate Thesis
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Project Purpose

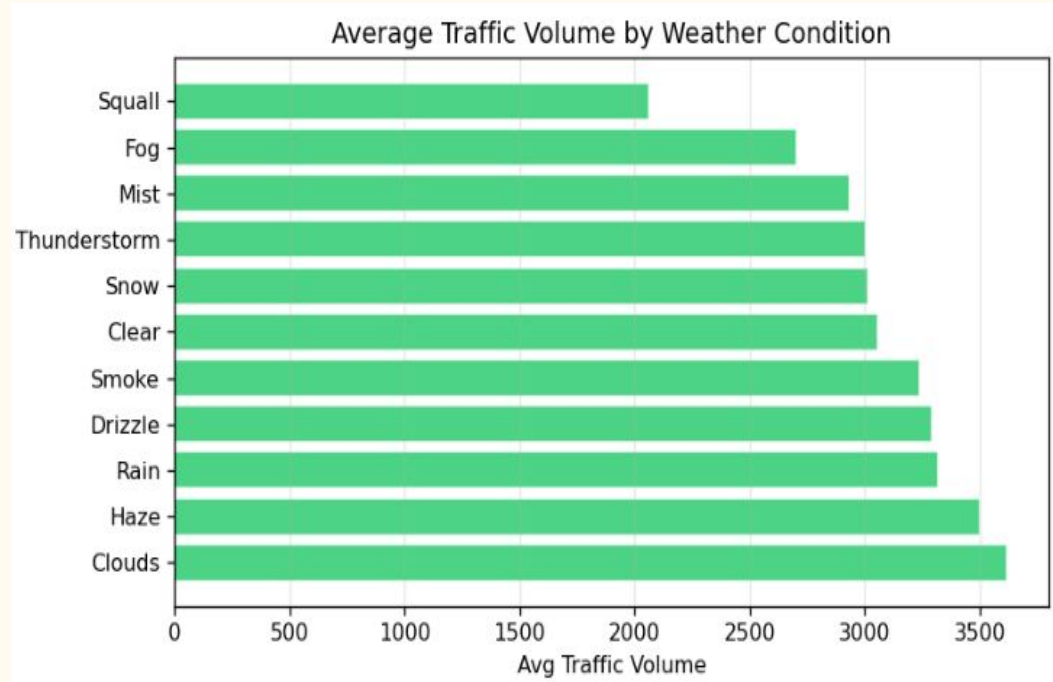
As cities grow, traditional traffic management systems often struggle to keep up with complex traffic patterns. Predicting urban traffic is particularly challenging because it depends on many dynamic factors like weather, time of day, and unexpected events, which do not easily fit into older and static models. The main goal of this project is to build a system that can accurately forecast urban traffic volume using machine learning algorithms.

This project tries to solve this issue by turning historical traffic data into predictive models. This approach is important because it helps cities become smarter and promotes open control systems.

Understanding Temporal Cycles



Environmental Impact on Flow



Applied Machine Learning Models and Methods

Model

Key Hyperparameters

Linear Regression

None

Random Forest

n_estimators=200, max_depth=15,
min_samples_split=5

XGBoost

n_estimators=300, lr=0.05, max_depth=8,
subsample=0.8

LightGBM

n_estimators=300, lr=0.05,
max_depth=8, num_leaves=63

SVR

kernel=rbf, C=100, epsilon=50
(trained on 8,000 samples)

Analysis of the Random Forest Algorithm

Random Forest was selected for its stability and resistance to overfitting. With an R2 of 0.9561, it successfully captures non-linear traffic patterns.

Evaluation Metric	Achieved Score
RMSE	417.2
MAE	232.2
R-squared (R2)	0.9561
CV RMSE (Mean ± Std)	433.5 ± 8.8
Training Time	5.44

Analysis of the XGBoost Algorithm

XGBoost delivered the study's peak performance with an R2 of 0.9646. Its gradient boosting framework excels at minimizing residuals, providing superior accuracy for peak-hour predictions where temporal changes are highest and error margins are critical.

Evaluation Metric	Achieved Score
RMSE	374.6
MAE	211.1
R-squared (R2)	0.9646
CV RMSE (Mean ± Std)	390.3 ± 9.3
Training Time	2.04

Analysis of Alternative and Baseline Models

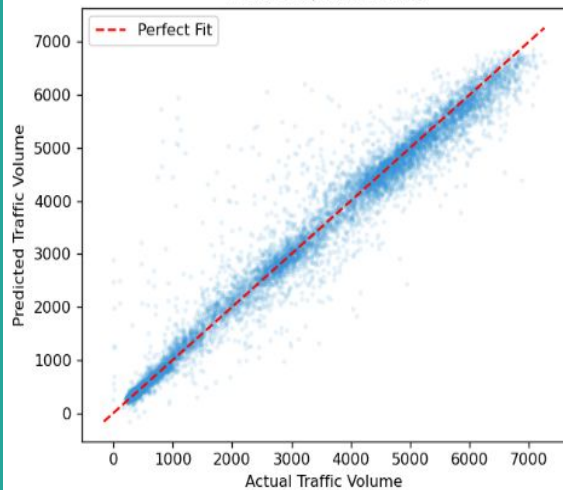
Revealing that simple linear assumptions are incapable for complex urban flows ($R^2=0.77$). In contrast, LightGBM offers exceptional efficiency, matching high-tier accuracy while increasing training time, making it the ideal candidate for real-time edge deployment.

Model Name	RMSE Score	R-squared (R^2)	Training Time
Linear Regression	943.5	0.7755	0.21
Support Vector Regression	924	0.7845	2.84
LightGBM	393	0.9604	1.49

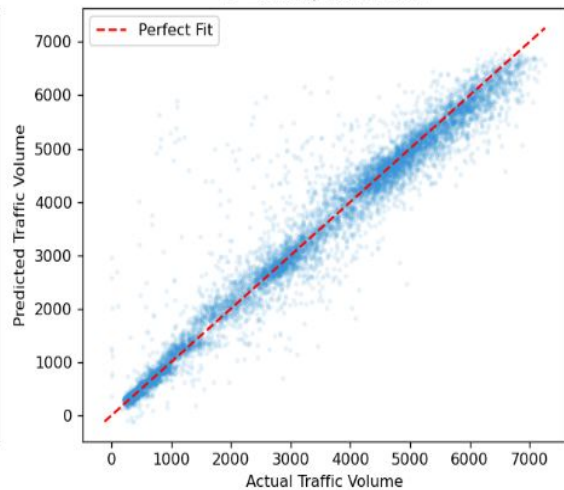
PREDICTION CONSISTENCY

(TOP 3 MODELS)

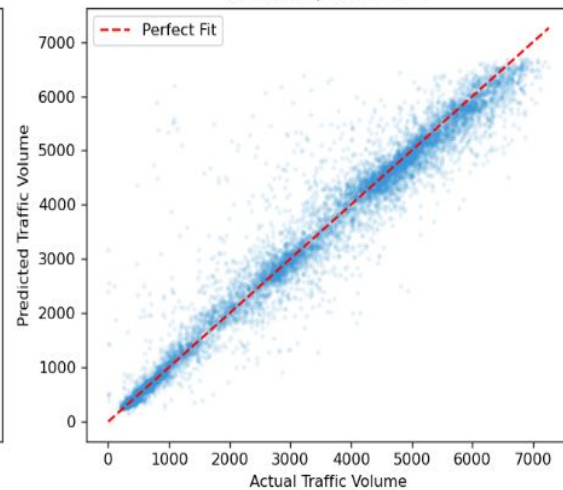
XGBoost
 $R^2=0.965$, RMSE=375



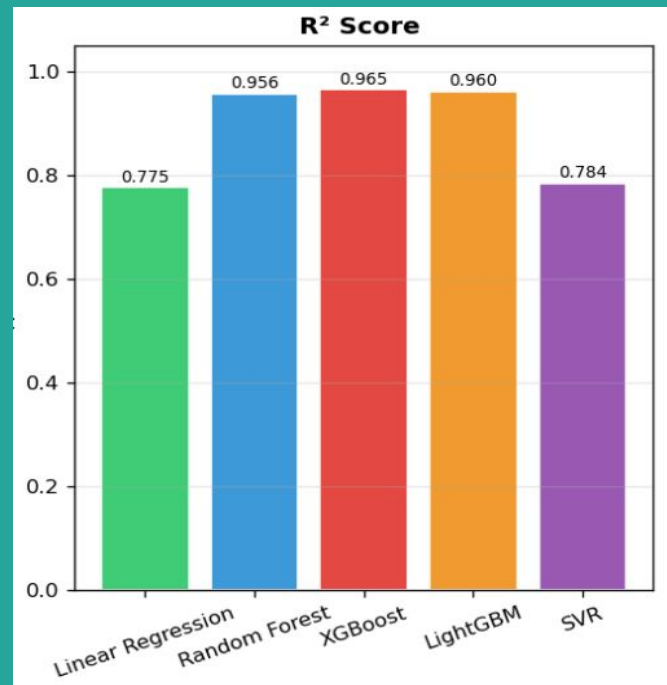
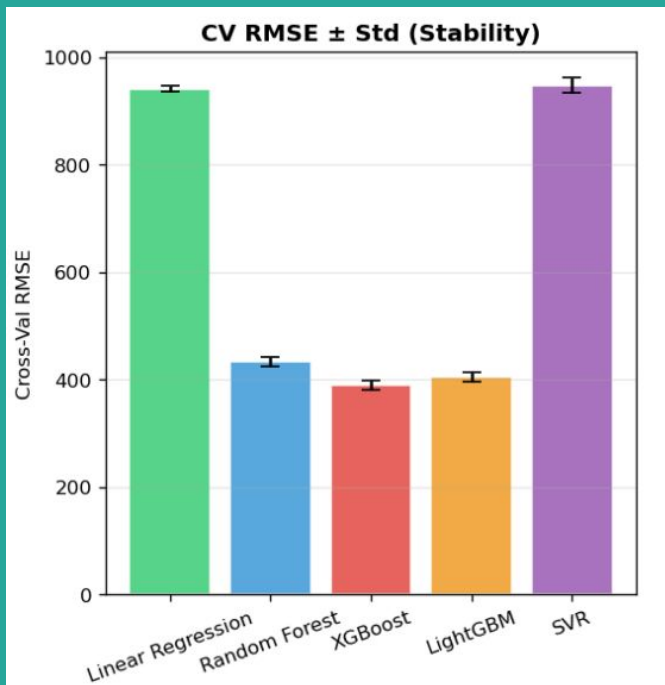
LightGBM
 $R^2=0.960$, RMSE=396



Random Forest
 $R^2=0.956$, RMSE=417



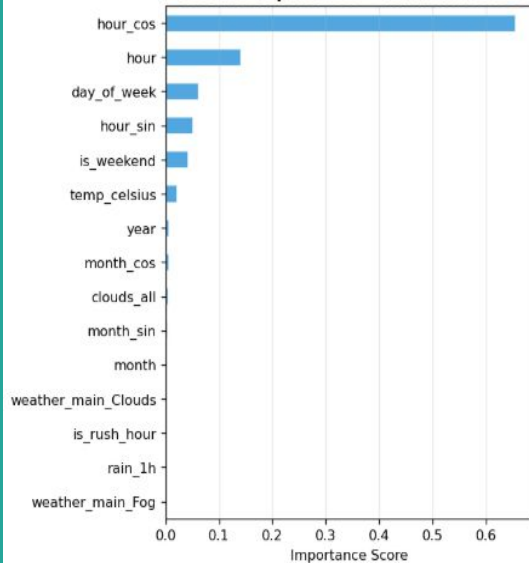
R² SCORE & STABILITY COMPRATION



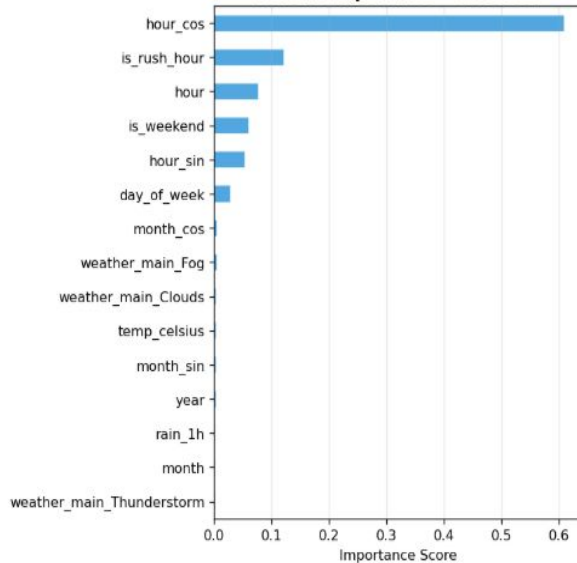
FEATURE IMPORTANCE

Top 15 Feature Importances (Tree-Based Models)

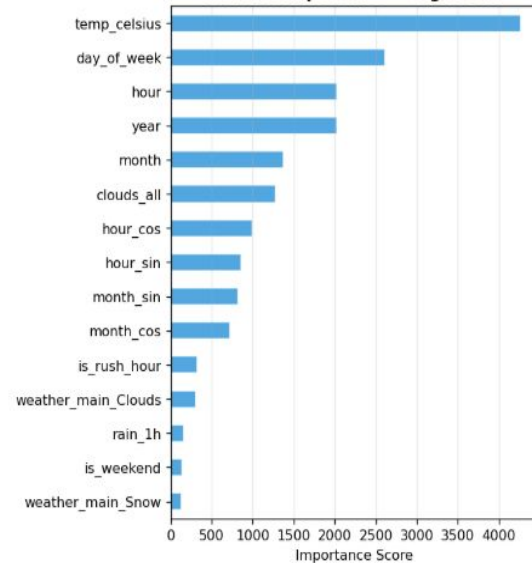
Feature Importance — Random Forest



Feature Importance — XGBoost



Feature Importance — LightGBM



Conclusion

Ensemble methods, particularly XGBoost, demonstrate exceptional accuracy ($R^2 = 0.965$) in capturing non-linear traffic patterns, establishing it as the optimal model for offline urban mobility forecasting. Also LightGBM shows predictive accuracy comparable to XGBoost while had faster training times; however, Linear Regression emerged as the most computationally efficient model.

THANK YOU.

question&discussion

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